Informatica Data Cleaning Task Report

1. Objective

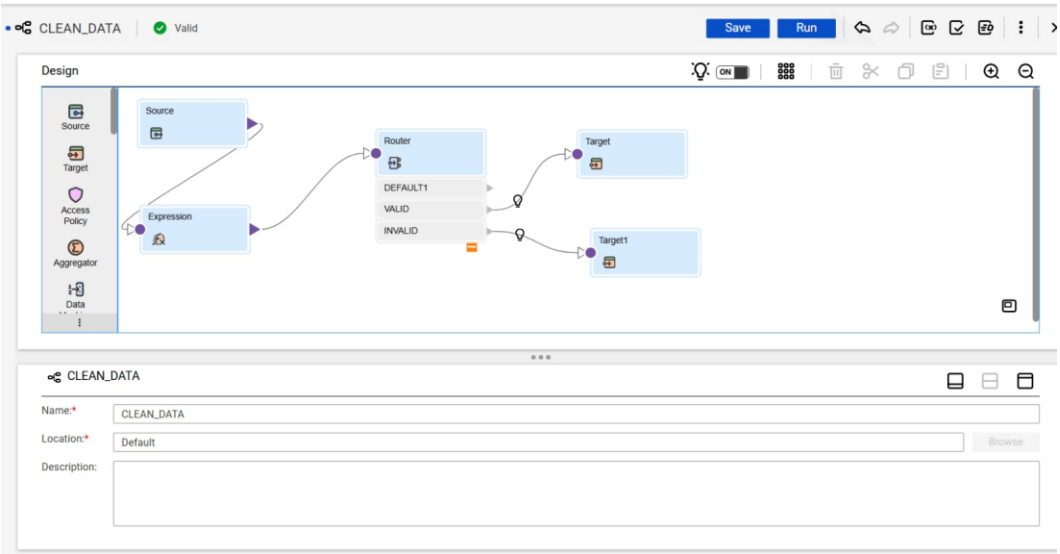
To create a data pipeline in Informatica to clean dirty sales transaction data based on business rules and load valid records into target while routing invalid records into a reject file.

2. Business Rules Applied

- Quantity cannot be negative – convert to positive.
- Date format must be in sync (YYYY-MM-DD).
- Null customers should go into rejected records.
- Null price should go to rejected records.
- Discount > 100% should go to rejected records.
- Null customer id should go in reject record.
- Zero quantity should go in rejected record.
- Negative discount should be changed to positive.

3. Mapping Design

Source → Expression (Data Cleaning) → Router (Valid / Invalid Split) → Valid Target & Reject Target.



4. Expression Transformation Logic

Data type validation using IS_NUMBER and IS_DATE functions. Conversion performed using TO_INTEGER, TO_DECIMAL, and TO_DATE. Negative values converted to positive using

conditional IIF logic.

Create simple expressions. You can also use expression macros to create complex expressions.

☐ Allow additional fields and expressions during task creation

Expressions

Field Name	Expression	Default Value	Type	Precision	Scale	Field Description	Link Path
C_CUSTOMER_ID	IIF(IS_NUMBER(CUSTOMER_ID), TO_	ERROR('transformation error')	integer	10	0		
C_PRODUCT_ID	IIF(IS_NUMBER(PRODUCT_ID), TO_I	ERROR('transformation error')	integer	10	0		
C_QUANTITY	IIF(IS_NUMBER(QUANTITY), IIF(TO_	ERROR('transformation error')	integer	10	0		
C_UNIT_PRICE	IIF(IS_NUMBER(UNIT_PRICE), IIF(TO	ERROR('transformation error')	decimal	10	0		
C_DISCOUNT	IIF(IS_NUMBER(DISCOUNT_PERCEN	ERROR('transformation error')	integer	10	0		
C_TRANSACTION_DATE	IIF(IS_DATE(TRANSACTION_DATE,'D	ERROR('transformation error')	date/time	29	9		
C_TRANSACTION_ID	IIF(IS_NUMBER(TRANSACTION_ID),	ERROR('transformation error')	integer	10	0		

5. Router Conditions

VALID Condition:

NOT ISNULL(C_CUSTOMER_ID)
AND NOT ISNULL(C_UNIT_PRICE)
AND C_QUANTITY > 0
AND C_DISCOUNT <= 100
AND NOT ISNULL(C_TRANSACTION_DATE)

INVALID Condition:

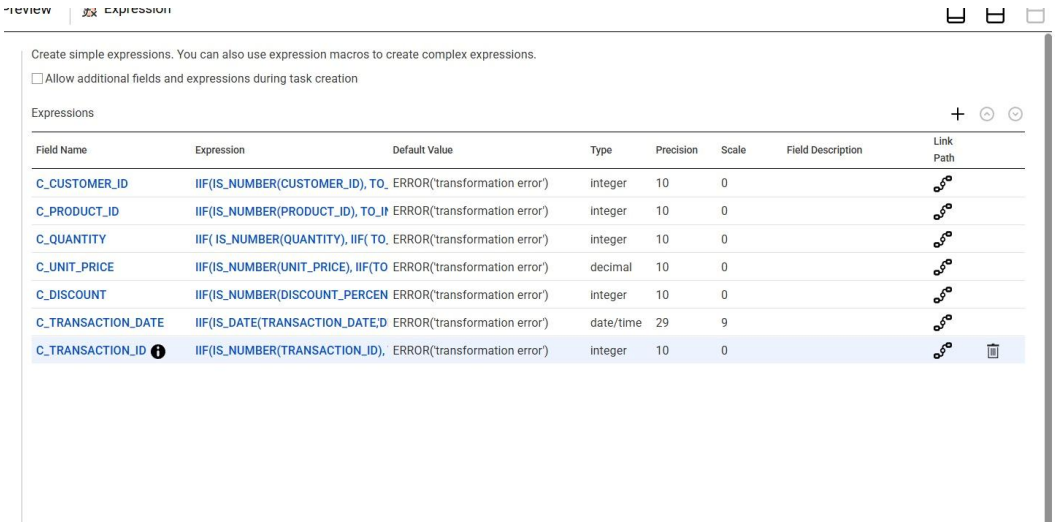
ISNULL(C_CUSTOMER_ID)
OR ISNULL(C_UNIT_PRICE)
OR ISNULL(C_QUANTITY)
OR C_QUANTITY = 0
OR C_DISCOUNT > 100
OR ISNULL(C_TRANSACTION_DATE)

Output Groups

Group Name	Condition
DEFAULT1	
VALID	NOT ISNULL(C_CUSTOMER_ID) AND NOT ISNULL(C_UNIT_PRICE) AND C_QUANTITY > 0 AND C_DISCOUNT <= 100 AND NOT ISNULL(C_TRANSACTION_DATE)
INVALID	ISNULL(C_CUSTOMER_ID) OR ISNULL(C_UNIT_PRICE) OR ISNULL(C_QUANTITY) OR C_QUANTITY = 0 OR C_DISCOUNT > 100 OR ISNULL(C_TRANSACTION_DATE)

6. Target Configuration

Valid records are written to main target table/file.
Invalid records are written to reject_hcl.csv.



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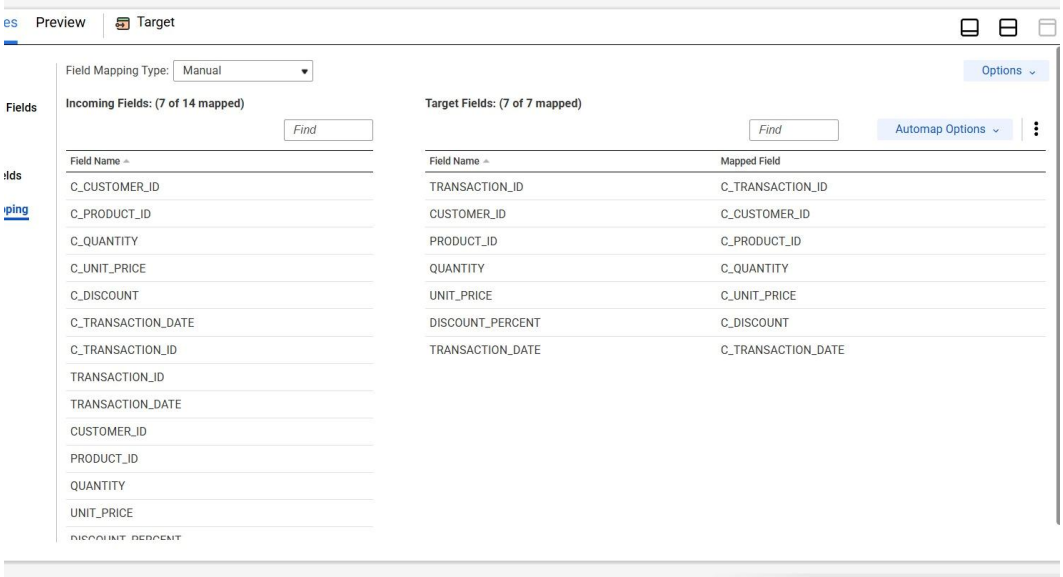

Preview | Target

Field Mapping Type: Manual

Options

Incoming Fields: (7 of 14 mapped)

Find

Field Name
C_CUSTOMER_ID
C_PRODUCT_ID
C_QUANTITY
C_UNIT_PRICE
C_DISCOUNT
C_TRANSACTION_DATE
C_TRANSACTION_ID
TRANSACTION_ID
TRANSACTION_DATE
CUSTOMER_ID
PRODUCT_ID
QUANTITY
UNIT_PRICE

Target Fields: (7 of 7 mapped)

Find

Automap Options

Field Name	Mapped Field
TRANSACTION_ID	C_TRANSACTION_ID
CUSTOMER_ID	C_CUSTOMER_ID
PRODUCT_ID	C_PRODUCT_ID
QUANTITY	C_QUANTITY
UNIT_PRICE	C_UNIT_PRICE
DISCOUNT_PERCENT	C_DISCOUNT
TRANSACTION_DATE	C_TRANSACTION_DATE

Details

Connection:* FLAT_FILE_CONN (Flat File) View... New Connection... New Parameter...

Object Type:* Single Object

Use a local file for target metadata

Object:* reject_hcl.csv Select... Formatting Options...

Operation: Insert

Advanced